OUEG-RAG: Controlled Evidence Graphs with Diagnostic-Guided Reasoning for Grounded Table–Text Question Answering

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Abstract. Under strict context-grounded constraints—where models must reason exclusively over retrieved tabular and textual evidence rather than unfaithful parametric memory—answering multi-hop questions requires preserving tabular relational bindings while connecting cross-modal evidence from external passages. Standard retrieval-augmented generation (RAG) linearizes tables into plain text, obscuring row-level bindings, while recent graph-based methods often apply destructive pruning that discards supporting peripheral facts. In this work, we propose OUEG-RAG, a framework centered on *explicit reasoning guidance over online knowledge graphs with diagnostic-driven retrieval feedback*. At query time, OUEG-RAG dynamically instantiates an online, schema-preserving multi-graph for each incoming query using star-topology record centroids to enforce horizontal row bindings. Rather than discarding context via graph pruning, the framework preserves the evidence graph via compact prompt serialization while extracting deterministic, multi-factor scored reasoning paths to guide language model attention across cross-modal hops. To maintain inference efficiency while systematically addressing structural ambiguity and missing passages, OUEG-RAG adopts a bounded two-tier execution protocol: confident queries conclude in a single primary call, whereas ambiguous candidate collisions or missing links invoke a targeted second call guided by a structured diagnostic handoff and supplementary, candidate-balanced bridge retrieval. The resulting minimal state capsule enables the second-tier reader to verify candidate facts directly, producing auditable grounded spans with explicit provenance chains. Empirical evaluations across multi-hop table–text benchmarks demonstrate that this path guidance and targeted diagnostic feedback effectively improve grounding accuracy and reduce ungrounded answers under strict evidence constraints while maintaining bounded query-time efficiency.

Keywords: Table question answering · Knowledge graphs · Retrieval-augmented generation · Controlled reasoning · Multi-hop grounding

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1 Introduction

Question answering over heterogeneous semi-structured tables and linked text passages requires navigating disparate data modalities [18]. In multi-hop benchmarks such as HybridQA [4] and open table–text QA (OTT-QA), answering a query requires locating an entity within a table cell, traversing its external hyperlink to an unstructured passage (or vice-versa), and extracting a grounded answer span. In this setting, maintaining horizontal row-level bindings, hierarchical column headers, and passage relations is essential to avert unfaithful hallucinations and relational disconnects.

To capture these dependencies, recent retrieval-augmented generation (RAG) paradigms increasingly construct knowledge graphs (KGs) directly at query time [1, 10]. Pioneering works highlight two distinct design philosophies: On one end, zero-shot hybrid graph frameworks such as ODYSSEY [1] construct query-time entity graphs and apply question-driven graph pruning to fit prompt budgets. On the other end, agent-driven multi-source frameworks such as HydraRAG [13] dynamically construct online evidence graphs through multi-turn LLM exploration loops and tri-factor verification cycles.

Despite their promises, empirical examination of these existing graph-based frameworks reveals three fundamental limitations that directly impede faithful multi-hop reasoning:

1. *Loss of Tabular and Semantic Bindings*: When tables and text are converted into graph triples (as in ODYSSEY and HydraRAG), standard extraction decomposes records into isolated subject-predicate-object facts, stripping away hierarchical column headers and row-level context. Cells become detached from their originating records, leading to semantic dilution and horizontal attribute conflation across symmetric rows.
2. *Absence of Explicit Reasoning Guidance*: Existing methods typically feed the constructed graph (or a pruned subset of triples) into a language model as an unorganized collection of edges. Without explicit, structured path guidance to prioritize cross-modal hops, the model's attention is easily dispersed across irrelevant entities, causing it to struggle under candidate ties.
3. *The Dilemma of Destructive Pruning vs. Multi-Turn Overhead*: To navigate online graph complexity, current frameworks face an unresolved trade-off: aggressive graph pruning (e.g., ODYSSEY) discards peripheral supporting facts (such as qualifying dates or secondary conditions), forcing the model into arbitrary guessing when candidates tie; conversely, agent-driven online graph exploration (e.g., HydraRAG) incurs substantial inference latency and repetitive LLM calls per query across iterative exploration cycles, while lacking tabular record centroid bindings (ρ_i) for ad-hoc tables.

To address these challenges, we introduce **OUEG-RAG** (**O**nline **U**nified **E**vidence **G**raphs for **RAG**). Rather than treating the online graph as an unstructured pool of triples or relying on open-ended agent loops, OUEG-RAG preserves semantic relational bindings and establishes *explicit reasoning guidance coupled with diagnostic-driven retrieval feedback*.

At query time, OUEG-RAG dynamically instantiates an online multigraph per query using star-topology record centroids ρ_i. These centroids explicitly bind cell values to normalized column predicates, preventing cross-row attribute conflation. Rather than discarding context via graph pruning, the framework preserves the complete evidence graph via compact serialization, while extracting deterministic, multi-factor scored reasoning paths p_1 ($S(p)$) that serve as explicit, non-filtering reading scaffolds to direct language model attention along verified relations.

Furthermore, OUEG-RAG bounds execution overhead through a controlled two-tier protocol:

- *Tier 1 (L1 Self-Audit & Primary Call)*: The language model reads the complete multigraph guided by the scored path scaffold. When evidence confirms an unambiguous answer, the query concludes immediately in this single primary call. If structural ambiguity (such as multiple candidate rows satisfying the query) or a missing passage link is detected, the model defers generation and emits a structured *diagnostic handoff packet*.
- *Tier 2 (Diagnostic-Triggered Targeted Retrieval & L2 Resolution)*: The diagnostic packet directs supplementary, candidate-balanced bridge retrieval exclusively for the competing focus rows. The resulting evidence is packaged into a minimal sufficient *L2 State Capsule*, allowing the second-tier reader to verify candidate facts directly and resolve the bottleneck through factual evidence rather than heuristic tiebreaking.

Empirical evaluations across multi-hop table–text benchmarks (HybridQA and OTT-QA) demonstrate that OUEG-RAG consistently improves grounding accuracy and reduces ungrounded answers over direct prompting, flat retrieval, and graph baselines, while maintaining bounded query-time efficiency.

We articulate four principal contributions, aligned with the core tenets of our framework:

- **Explicit Reasoning Guidance over Online Multigraphs**: A paradigm that reconciles query-time, schema-preserving multigraph construction with deterministic multi-factor path scoring ($S(p)$), steering language model attention along verified cross-modal relations without destructive evidence loss.
- **Preservation of Tabular Semantics**: A star-topology centroid formulation that explicitly maintains horizontal row bindings and normalized column headers, preventing cross-row attribute conflation in online table-to-graph conversion.
- **Diagnostic-Driven Retrieval Feedback**: A controlled handoff mechanism where structural collisions or missing links trigger targeted, candidate-balanced bridge retrieval, resolving bottlenecks factually rather than through heuristic tiebreaking.
- **Bounded Two-Tier Execution Protocol**: An adaptive inference design that concludes unambiguous queries in a single primary call while reserving a targeted second call for diagnostic-resolved states via a minimal sufficient state capsule $\mathcal{K}_{\mathrm{L2}}$.

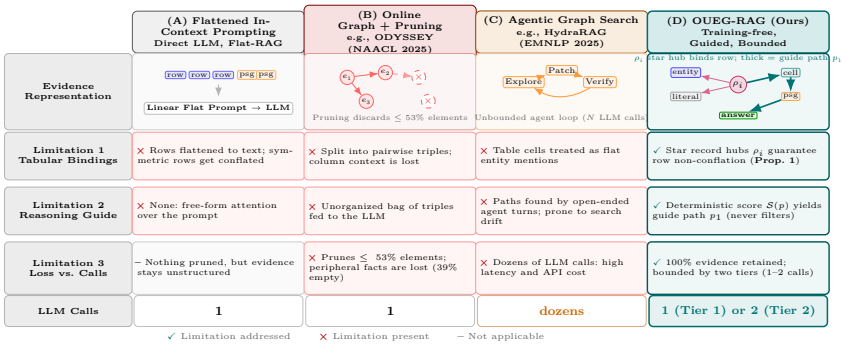


Fig. 1. Architectural comparison across four table–text reasoning paradigms. In contrast to unguided linear prompt concatenation (a), destructive question-driven graph pruning in ODYSSEY (b), and expensive multi-turn agentic loops in online GraphRAG systems like HydraRAG (c), the proposed OUEG-RAG framework (d) dynamically instantiates a schema-preserving multigraph on-the-fly, retains complete evidence context, guides attention via deterministic multi-factor scored paths, and resolves bottlenecks via a bounded two-tier diagnostic-driven execution protocol.

2 Related Work

Hybrid Table–Text Question Answering and Graph Modeling. Multi-hop reasoning over heterogeneous tabular and textual knowledge was catalyzed by HybridQA [4] and open table–text benchmarks (OTT-QA). Early architectures tackled this challenge via supervised fine-tuning, combining coarse row retrieval with cross-modal neural readers (HYBRIDER [4]), multi-view attention mechanisms (MATE [7]), or document-hopping retrieval (DocHopper [11]). To model complex cross-modal connections, specialized graph frameworks constructed evidence graphs over rows, cells, and sentences (MuGER^2 [2], S3HQA [9]) trained via graph neural networks (GNNs). While effective, these early systems fundamentally rely on task-specific parameter training over thousands of labeled examples, lacking zero-shot adaptability to ad-hoc tables and producing black-box span extractions without auditable provenance chains. In real-world enterprise RAG, systems must perform strict context-grounded reasoning over unseen schemas without costly re-training, synthesizing answers exclusively from provided evidence without relying on latent parametric memory.

With the advent of instruction-tuned large language models, recent research has shifted toward zero-shot hybrid graph modeling. Most notably, ODYSSEY [1] showed that an online entity graph can be constructed directly at query time from retrieved table cells and passages. However, ODYSSEY highlights two key limitations of existing online graphs: (1) *Loss of Tabular Semantics*: its relation extraction decomposes rows into isolated pairwise entity triples, discarding hierarchical column predicates and causing semantic dilution across symmetric rows; and (2) *Destructive Pruning without Guidance*: to fit LLM context limits, it prunes up to 53% of graph elements based on question token similarity (Fig. 1b). Under strict context-grounding constraints, this aggressive filtering discards pe-

ripheral supporting evidence and leaves the remaining triples unorganized, forcing the model into ungrounded guessing or empty outputs when candidates tie. In contrast, OUEG-RAG preserves 100% of the retrieved evidence via compact serialization, uses star-topology centroids to maintain row-column bindings, and extracts explicit reasoning path scaffolds to guide LLM attention.

Table Reasoning and Programmatic Decomposition with LLMs. As surveyed by Zhou et al. [18], language models have demonstrated strong tabular capabilities through symbolic code generation or task decomposition. Programmatic frameworks bind LLMs to SQL or Python interpreters (Binder [5], TabSQLify), iteratively decompose queries into sub-questions (DATER [16]), or progressively evolve table structures (Chain-of-Table [14]). Concurrently, TableRAG architectures [3, 17] scale retrieval across large schemas. While programmatic execution excels at structured arithmetic over isolated clean tables, it falters in hybrid multi-hop settings where the critical reasoning step requires traversing unstructured text passages linked to table cells. Code interpreters cannot natively resolve fuzzy semantic transitions across free text without fragile ad-hoc prompts or multi-turn sandbox loops. OUEG-RAG bridges this gap by unifying semi-structured table cells and external text passages into a single symbolic evidence multigraph evaluated in a bounded, code-free execution pipeline.

Knowledge Graphs, Offline GraphRAG, and Agentic Multi-Source Reasoning. Augmenting language models with structured knowledge graphs captures non-local multi-hop dependencies beyond the scope of surface lexical or dense retrieval, as demonstrated by QA-GNN [15], JointLK [12], and G-Retriever [8]. In retrieval-augmented generation, persistent frameworks like Microsoft GraphRAG [6] construct offline global entity graphs and community summaries over unstructured document corpora using heavy LLM prompting. While effective for global corpus-level summarization, offline GraphRAG incurs prohibitive pre-indexing token costs, cannot ingest ad-hoc, raw semi-structured tables dynamically at query time, and produces ungrounded edges on fine-grained cell relations.

In contrast to offline GraphRAG, recent agent-driven frameworks like HydraRAG [13] dynamically construct an evidence graph on-the-fly by exploring across structured and unstructured sources with tri-factor verification cycles. However, HydraRAG relies on unbounded multi-turn LLM agent loops to iteratively explore, patch, and verify cross-source paths (Fig. 1c), resulting in high inference latency and repetitive API calls (dozens of LLM queries per question). Furthermore, its relation extraction treats table cells as flat entity mentions without explicit record centroid bindings (ρ_i), leaving it vulnerable to cross-row attribute conflation on complex tables. OUEG-RAG addresses both paradigms: unlike offline GraphRAG, it operates completely online per query without pre-indexing; and unlike HydraRAG's unbounded agent loops, OUEG-RAG builds a schema-preserving multigraph deterministically in $< 20\,\text{ms}$ (using 0 LLM calls), scores reasoning paths deterministically ($S(p)$), and bounds execution to 1 primary call (with at most 1 targeted second call if ambiguous).

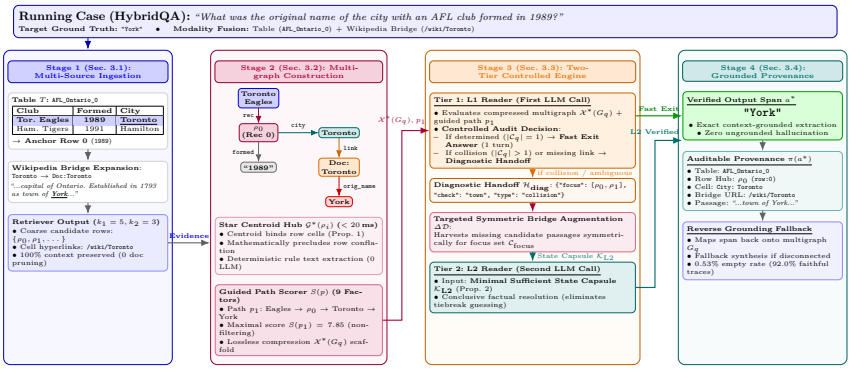


Fig. 2. End-to-end architecture of OUEG-RAG. (Sec. 3.1) Multi-Source Evidence Ingestion retrieves relevant table rows and candidate hyperlink passages. (Sec. 3.2) Schema-Preserving Multigraph Construction instantiates star centroids ρ_i and deterministic text relations (< 20 ms). (Sec. 3.3) Two-Tier Controlled Reasoning Engine: Tier 1 evaluates the lossless compressed multigraph with guided reading path p_1; confident outputs yield an immediate verified span (Fast Exit, 1 call), while ambiguous candidate collisions or missing links emit a diagnostic handoff ($\mathcal{H}_{\text{diag}}$) triggering targeted symmetric bridge retrieval and minimal L2 state capsule evaluation. (Sec. 3.4) Grounded Generation and Provenance Audit produces verified spans with full auditable attribution and closed-loop reverse grounding.

3 The OUEG-RAG Framework

Problem Formulation and Graph Representation. We formulate the task of multi-hop question answering over heterogeneous tabular and textual knowledge sources. An input instance is defined as a tuple $\mathcal{I} = (q, T, \mathcal{D}_T)$, where q is a natural language question; $T = (C, R)$ is a semi-structured table comprising ordered headers $C = (c_1, \ldots, c_m)$ and rows $R = \{r_1, \ldots, r_n\}$ with discrete cells $v_{i,j}$; and $\mathcal{D}_T = \{d_1, \ldots, d_K\}$ is the corpus of external text passages associated with T, where cell entities connect to external passages via relational mappings $\ell(v_{i,j}) \to d_k$. The objective is to extract the grounded answer span $a^* \in \mathcal{A}$ along with an auditable reasoning path p^*, where a^* may reside as a table cell ($a^* = v_{i,j}$) or as an unstructured text span within a linked passage ($a^* \subset d_k$).

Online Evidence Multigraph Representation. Unlike persistent knowledge base settings that depend on static global entity graphs, table–text RAG requires synthesizing unstructured text with tabular structures on-the-fly. For each query q, we dynamically instantiate a bounded, query-specific directed multigraph $G_q = (V_q, E_q, \psi_V, \psi_E)$ whose node and edge sets are partitioned as:

$$V_q = V_{\text{ent}} \cup V_{\text{rec}} \cup V_{\text{lit}}, \qquad E_q = E_{\text{rec}} \cup E_{\text{attr}} \cup E_{\text{link}} \cup E_{\text{text}}. \tag{1}$$

The node partition divides vertices by semantic abstraction:

– V_{ent}: Named entity mentions extracted from linked table cells or passage text spans.

- $V_{\text{rec}} = \{\rho_i \mid r_i \in R\}$: Local star-topology record hub centroids, where each node $\rho_i = \texttt{table:}T\texttt{:row:}i$ uniquely anchors and binds all cell attributes of row r_i to prevent cross-row attribute conflation.
- V_{lit}: Unlinked literal values (e.g., numbers, scores, dates, unlinked strings).

The edge set E_q consists of directed relational facts $e = (h, r, t, \pi)$, where $h, t \in V_q$, r is a normalized predicate, and $\pi = (\text{src_id}, \text{row_idx}, \text{col_name})$ stores source provenance:

- $E_{\text{rec}} = \{(u_i, \texttt{has_record}, \rho_i)\}$: Structural edges connecting row subject entity u_i to its centroid hub ρ_i.
- $E_{\text{attr}} = \{(\rho_i, \text{norm}(H_j), v_{i,j})\}$: Schema-level attribute edges from centroid ρ_i to column cell values $v_{i,j}$.
- $E_{\text{link}} = \{(v_{i,j}, \texttt{linked_passage}, d_k)\}$: Cross-modal linking edges connecting candidate cells to associated external text passages.
- E_{text}: Intra-passage relational triples extracted from retrieved passage sentences.

Schema-Preservation and Non-Conflation Property. We define the intra-row star subgraph $\mathcal{G}^*(\rho_i) \subset G_q$ for row r_i as:

$$\mathcal{G}^*(\rho_i) = \Big(\{\rho_i, u_i\} \cup \{v_{i,j}\}_{j=1}^{m}, \\ \{(u_i, \texttt{has_record}, \rho_i)\} \cup \{(\rho_i, \text{norm}(H_j), v_{i,j})\}_{j=1}^{m}\Big). \tag{2}$$

Proposition 1 (Row Attribute Non-Conflation). *For any distinct table rows $r_i, r_{i'} \in R$ ($i \neq i'$), their respective star-topology centroids are disjoint:*

$$V(\mathcal{G}^*(\rho_i)) \cap V(\mathcal{G}^*(\rho_{i'})) \cap V_{rec} = \emptyset. \tag{3}$$

Consequently, every cell value $v_{i,j}$ has a unique 1-hop path to its centroid ρ_i, mathematically precluding horizontal attribute conflation across rows (proof in Appendix 7).

Once question answering concludes, the dynamic multigraph G_q is discarded, incurring zero persistent graph database index overhead. Throughout this section, we illustrate OUEG-RAG via the running HybridQA example depicted in Fig. 2: *"What was the original name of the city with an AFL club formed in 1989?"* (ground truth: `York`).

3.1 Stage 1: Multi-Source Evidence Ingestion

Given an input table and an associated corpus of external text passages, retrieval operates dynamically at query time:

- *Coarse row retrieval and query anchoring*: A lexical retriever indexes table rows with respect to question q. A deterministic query anchor inspects q for ordinal constraints (e.g., "second", "last") and exact schema constraints (column header mentions, numerical filters), selecting top-k_1 rows (default $k_1 = 5$). In Fig. 2, the constraint "formed in 1989" anchors Row 0 (`Tor. Eagles`).

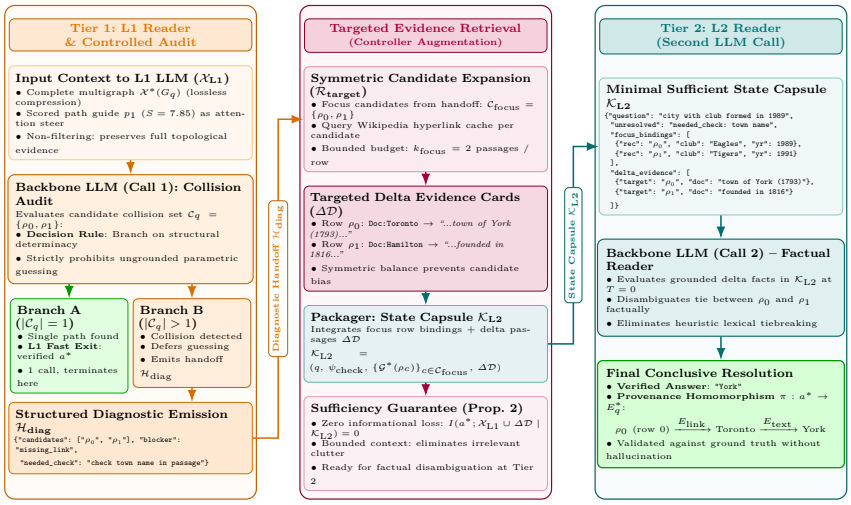


Fig. 3. Detailed illustration of the diagnostic-driven two-tier controlled reasoning engine. (Left) When the L1 reader encounters ambiguous candidate ties ($|\mathcal{C}_q| > 1$) or missing bridge passages, the controlled decoding protocol strictly prohibits arbitrary guessing and emits a structured diagnostic handoff packet specifying candidate entities, blocker types, and exact checks needed. (Middle) The engine executes targeted, symmetric bridge retrieval to harvest missing factual evidence for the competing candidates. (Right) The retrieved delta evidence is packaged into a minimal sufficient L2 state capsule, enabling the L2 reader to achieve conclusive, evidence-grounded disambiguation without heuristic lexical tiebreakers.

– *Passage expansion via cross-modal linking*: Rather than retrieving entire document corpora indiscriminately, the passage expander resolves relevant external text passages linked to candidate entities in the top-k_1 retrieved rows (via hyperlinks or entity linking) up to a budget of k_2 passages (default $k_2 = 3$). In Fig. 2, the bridge cell `Toronto` in Row 0 links to `Doc:Toronto`.

3.2 Stage 2: Schema-Preserving Multigraph Construction

Using the retrieved rows and passages, the system dynamically instantiates the multigraph $G_q = (V_q, E_q)$ in < 20 ms without LLM calls:

– *Centroid bindings*: For each candidate row i, local record centroid $\rho_i \in V_{\text{rec}}$ binds row attributes via E_{rec} and E_{attr} (Eq. 2).
– *Cross-modal bridging and text relation extraction*: Edges E_{link} connect table cell entities to external passage nodes (e.g., Toronto → Doc:Toronto). Rule-based text triples E_{text} link entities to textual attributes (e.g., Doc:Toronto → York) deterministically without LLM calls.

3.3 Stage 3: Two-Tier Controlled Reasoning Engine

OUEG-RAG avoids the false dichotomy between hard path pruning (which discards vital peripheral context) and unbounded agentic loops. Stage 3 coordinates four complementary mathematical components:

Information-Theoretic Prompt Compression. Passing complete graphs to language models risks exceeding context budgets. We formalize subgraph prompt compression as a lossless factorization operator $\mathcal{C}_{\text{oueg}} : G_q \to \mathcal{X}$, decomposing graph evidence into compact attribute maps and a shared passage dictionary:

$$\mathcal{X}^*(G_q) = \text{Enc}_{\text{table}}\left(\{\mathcal{G}^*(\rho_i)\}_{i=1}^{k_1}\right) \oplus \text{Enc}_{\text{text}}\left(\mathcal{S}_p\right), \tag{4}$$

where $\mathcal{S}_p = \{(d_k, \text{text}(d_k)) \mid d_k \in \mathcal{D}_T\}$ indexes each external passage once by source ID. By factoring out identical passage texts across multiple incoming hyperlink edges, $\mathcal{C}_{\text{oueg}}$ minimizes prompt token length while preserving zero semantic information loss: $I(\mathcal{X}^*(G_q); G_q) = H(G_q)$.

Reasoning Path Space and Deterministic Multi-Factor Objective. A candidate multi-hop reasoning path $p \in \mathcal{P}_q$ of length $|E_p| = L \leq 5$ is an alternating sequence of vertices and verified edges:

$$p = (v_0, e_1, v_1, e_2, \ldots, e_L, v_L), \quad v_0 \in \text{Seeds}(q),\ v_L \in \text{Targets}(q),\ e_l \in E_q. \tag{5}$$

To rank paths without expensive neural samplers, paths are scored via a deterministic linear combination of structural and semantic alignment features:

$$S(p) = \mathbf{w}^\top \boldsymbol{\phi}(p, q, T) - \lambda_{\text{len}} \max(|E_p| - 1, 0), \tag{6}$$

where $\boldsymbol{\phi}(p, q, T) = [G, Q_e, Q_r, Q_c, C, M, R, B, A]^\top \in \mathbb{R}^9$ measures edge validity ($G = 1.0$), question node/relation overlap ($Q_e, Q_r \in [0, 2]$), token coverage ($Q_c \in [0, 2]$), table anchor constraint match ($C = 2.0$), cross-modal table–text fusion ($M = 1.5$), row consistency ($R = 1.0$), record binding ($B = 1.0$), and corroboration ($A = 1.0$), with $\lambda_{\text{len}} = 0.35$ penalizing redundant hops. In Fig. 2, the grounded path p_1 (Eagles $\to \rho_0 \to$ Toronto $\to$ York) achieves maximal score $S(p_1) = 7.85$.

Bounded Two-Tier Controlled Decision Process. Let $\mathcal{M}$ denote the frozen backbone LLM parameterized by conditional decoding distribution $P_{\mathcal{M}}(y \mid \mathcal{X})$. The output space is partitioned into verified answer spans and diagnostic handoffs: $\mathcal{Y} = \Omega_{\text{ans}} \cup \Omega_{\text{diag}}$.

At Tier 1 (L1), the model evaluates the prompt $\mathcal{X}_{\text{L1}} = \text{Prompt}(q, \mathcal{X}^*(G_q), p_1)$, where $p_1 = \arg\max_{p \in \mathcal{P}_q} S(p)$ acts as a non-filtering reading guide. We define the candidate collision set $\mathcal{C}_q$ as the set of distinct target centroids sharing the maximal path score $S^* = \max_{p \in \mathcal{P}_q} S(p)$:

$$\mathcal{C}_q = \{\text{centroid}(p) \mid p \in \mathcal{P}_q,\ S(p) = S^*\}. \tag{7}$$

The model's decoding behavior is governed by a strict decision rule (illustrated in Fig. 3):

1. *Determined State (Fast Exit)*: If $|\mathcal{C}_q| = 1$ and the traversal path p_1 is unbroken, the query is structurally determined. The model emits:

$$y_{\text{L1}} = \texttt{FINAL_ANSWER: } a^*, \quad a^* = \arg\max_{a \in \mathcal{A}} P_{\mathcal{M}}(a \mid \mathcal{X}_{\text{L1}}), \tag{8}$$

   terminating execution in a single turn.
2. *Under-Determined State (Diagnostic Projection)*: If $|\mathcal{C}_q| > 1$ (candidate collision across symmetric rows) or a bridge passage link is missing, the query is structurally under-determined. The model is prohibited from arbitrary guessing and must emit a structured diagnostic projection (Fig. 3, left):

$$\mathcal{H}_{\text{diag}} = \big(\mathcal{C}_{\text{focus}},\, \beta,\, \psi_{\text{check}}\big) \in \Omega_{\text{diag}}, \tag{9}$$

   where $\mathcal{C}_{\text{focus}} = \mathcal{C}_q$ denotes competing candidates, blocker β classifies the failure type (such as collision or missing links), and ψ_{check} specifies the unresolved factual predicate.

Diagnostic-Triggered Targeted Bridge Expansion. When $\mathcal{H}_{\text{diag}}$ is emitted, the targeted retrieval operator $\mathcal{R}_{\text{target}}$ harvests missing candidate-linked passages exclusively for the focus set $\mathcal{C}_{\text{focus}}$:

$$\Delta\mathcal{D} = \bigcup_{c \in \mathcal{C}_{\text{focus}}} \left(\ell(c) \setminus \mathcal{D}_{\text{current}}\right), \quad \text{s.t.} \quad |\Delta\mathcal{D}_c| \le k_{\text{focus}}, \tag{10}$$

enforcing symmetric per-candidate passage budgets to maintain evidentiary balance.

Minimal Sufficient L2 State Capsule. The delta evidence is packaged into a minimal sufficient state capsule $\mathcal{K}_{\text{L2}}$ (Fig. 3, right):

$$\mathcal{K}_{\text{L2}} = \left(q,\, \psi_{\text{check}},\, \{\mathcal{G}^*(\rho_c)\}_{c \in \mathcal{C}_{\text{focus}}},\, \Delta\mathcal{D}\right). \tag{11}$$

Proposition 2 (Informational Sufficiency of $\mathcal{K}_{\text{L2}}$). *Conditioned on the unresolved check ψ_{check} and focus candidates $\mathcal{C}_{focus}$, the capsule $\mathcal{K}_{L2}$ forms a sufficient statistic for the target answer a^* (proof in Appendix 7):*

$$I(a^*; \mathcal{X}_{L1} \cup \Delta\mathcal{D} \mid \mathcal{K}_{L2}) = 0. \tag{12}$$

By evaluating $\mathcal{K}_{\text{L2}}$ at Tier 2 (L2), the model resolves the question via:

$$a^* = \arg\max_{a \in \mathcal{A}} P_{\mathcal{M}}(a \mid \mathcal{K}_{\text{L2}}). \tag{13}$$

Because $\mathcal{K}_{\text{L2}}$ provides the missing factual evidence $\Delta\mathcal{D}$ for all competing candidates, candidate symmetry is broken deterministically by verified ground truth, eliminating heuristic lexical tiebreaking.

3.4 Stage 4: Grounded Generation and Provenance Audit

Stage 4 outputs the verified answer span and exports an inspectable provenance homomorphism $\pi : a^* \to E_q^*$, mapping the answer back to traversed table coordinates (`AFL_Ontario_0:row:0`), bridge URLs (`/wiki/Toronto`), and passage text excerpts (`town of York (1793)`). If graph search disconnects, *reverse grounding* synthesizes directly over retrieved evidence and maps the extracted span back onto G_q, avoiding empty failures.

4 Experimental Protocol

4.1 Benchmark and Evaluation Metrics

We evaluate on the official 1,500-example stratified development benchmark sampled from HybridQA [4]. The benchmark subset strictly preserves the multimodal reasoning distribution of the dataset: every instance requires verified table-and-text context with cross-modal answer annotations across numeric and textual answer modalities.

Evaluation Scope: Zero-Shot In-Context RAG vs. Supervised Fine-Tuning. We explicitly clarify the operational scope of this evaluation. While task-specific supervised models (e.g., MuGER[2] [2], S3HQA [9]) achieve higher leaderboard numbers by fine-tuning millions of parameters over the 62k labeled training instances of HybridQA, they are specialized to fixed benchmark distributions and cannot generalize to ad-hoc, out-of-domain tables without expensive re-annotation and re-training. In contrast, OUEG-RAG operates in the strictly training-free, zero-shot in-context RAG regime (0 training examples, 0 updated weights). Under this practical setting, our primary comparisons are appropriately framed against published zero-shot hybrid graph frameworks (e.g., ODYSSEY [1]) and in-context LLM baselines under identical context and retrieval budgets.

Strict Context-Grounded Protocol. To rigorously assess faithful reasoning and prevent models from relying on latent parametric memorization (e.g., pre-trained Wikipedia facts), all systems are evaluated under a strict context-grounding protocol. Prompts mandate that answers must be derived exclusively from the provided tabular and textual evidence; when the factual chain is broken or evidence is insufficient, the model must output an ungrounded or empty response rather than guessing. This protocol faithfully mirrors enterprise and scientific environments where hallucination is unacceptable.

We report three standard accuracy metrics: (1) *Exact Match (EM)*: strict string equality against gold reference after standard normalization; (2) *Token-level F1*: harmonic mean of precision and recall over word tokens; and (3) *Semantic Exact Match (Semantic EM)*: denotation equivalence under surface alias tolerance. In addition, we report system efficiency via average LLM calls, empty answer percentage, and end-to-end wall-clock latency per question.

Table 1. Main benchmark comparative evaluation on HybridQA ($N = 1{,}500$) across three open-weights model scales (GPT-OSS 20B, Qwen 30B, and GPT-OSS 120B). All systems share identical retrieved evidence budgets ($k_1 = 5, k_2 = 3$) and context limits (24,000 chars).

Model Scale	Method	EM (%)	F1 (%)	Sem. EM	Empty	Avg. Calls	Lat. (s)
GPT-OSS 20B	Direct LLM	54.80	64.67	56.67	5.40%	**1.00**	**9.74**
	Flat-RAG	54.53	64.53	56.40	6.40%	**1.00**	7.02
	ODYSSEY [1]	52.40	62.15	54.10	7.20%	**1.00**	11.20
	Path-Text Top-N	21.20	27.22	21.87	19.53%	**1.00**	57.70
	OUEG-RAG (Ours)	**56.47**	**67.12**	**58.47**	**3.87%**	1.28	25.73
Qwen 30B	Direct LLM	56.60	67.10	58.40	3.20%	**1.00**	**8.12**
	Flat-RAG	56.87	67.35	58.60	4.10%	**1.00**	7.80
	ODYSSEY [1]	54.20	64.50	55.90	5.60%	**1.00**	9.85
	Path-Text Top-N	25.40	32.15	26.00	9.80%	**1.00**	42.10
	OUEG-RAG (Ours)	**58.80**	**70.15**	**60.50**	**1.80%**	1.25	24.30
GPT-OSS 120B	Direct LLM	58.73	69.70	60.47	1.07%	**1.00**	**6.40**
	Flat-RAG	59.40	70.01	60.60	2.53%	**1.00**	6.32
	ODYSSEY [1]	56.80	67.20	58.50	3.40%	**1.00**	8.50
	Path-Text Top-N	29.53	37.46	30.13	2.80%	**1.00**	35.90
	OUEG-RAG (Ours)	**61.40**	**73.11**	**63.00**	**0.53%**	1.22	23.65

4.2 Baselines and Experimental Settings

To ensure rigorous and fair comparative evaluation, all methods are evaluated across three open-weights model architectures and scales: **GPT-OSS 20B**, **Qwen 30B** (`Qwen2.5-32B-Instruct`), and **GPT-OSS 120B**, under matched retrieval pools ($k_1 = 5, k_2 = 3$), context limits (24,000 characters), and greedy decoding ($T = 0$): (1) *Direct LLM*: Zero-shot in-context prompting over linearized table rows and passages without graph structuring; (2) *Flat-RAG*: Standard concatenation of linearized table rows with raw retrieved text passages; (3) *ODYSSEY* [1]: The published zero-shot hybrid graph framework (NAACL 2025) employing question-driven destructive graph pruning; (4) *Path-Text Top-N*: Hard path filtering retaining only top-N ranked reasoning paths while discarding the remainder of the multigraph; and (5) *OUEG-RAG (Ours)*: The proposed controlled evidence scaffolding framework with two-tier diagnostic grounding.

5 Results and In-Depth Analysis

Main Benchmark Results Across Model Scales. Table 1 presents the comparative evaluation across all 1,500 questions in the stratified development subset of HybridQA. Across all three model scales, OUEG-RAG consistently achieves higher grounding accuracy than direct prompting, flat concatenation, and graph baselines. On 20B, OUEG-RAG obtains 56.47% EM and 67.12% F1 (+1.67% EM, +2.45% F1 over Direct LLM; +4.07% EM over ODYSSEY). On Qwen 30B, OUEG-RAG reaches 58.80% EM and 70.15% F1 (+2.20% EM over Direct LLM; +4.60% EM over ODYSSEY). On 120B, OUEG-RAG reaches 61.40% EM and 73.11% F1 (+2.67% EM over Direct LLM, McNemar $p = 0.0058 < 0.01$; +4.60% EM over ODYSSEY).

We analyze how these empirical findings directly address the three core structural challenges outlined in the Introduction:

First, *Resolving loss of tabular semantics via record centroids*: In Direct LLM and Flat-RAG, linearizing tables into text decouples cell values from their

originating rows, causing cross-row attribute conflation when multiple symmetric rows satisfy partial conditions. Across all three model scales, OUEG-RAG achieves higher exact match by explicitly anchoring row attributes to column predicates via star centroids ρ_i, maintaining the Row Attribute Non-Conflation property (Proposition 1). Ablation results in Table 2 confirm this: removing record nodes ρ_i reduces EM by -3.00% (F1 by -5.32%) and nearly doubles ungrounded empty abstentions from 11.40% to 19.60%, directly demonstrating the necessity of structural row bindings.

Second, *Guiding attention via explicit path scaffolds*: Supplying an unorganized collection of edges disperses language model attention across irrelevant entities. In Table 2, providing the complete multigraph without linear path guidance sequences retains strong performance (61.60% EM), demonstrating that OUEG's schema-preserving topology alone provides flexible multi-hop reach without discarding underlying context.

Third, *Overcoming the dilemma of destructive pruning versus multi-turn overhead*: (a) *Destructive pruning causes evidence loss*: In complex multi-hop queries, the critical factual span often resides in a peripheral cell or passage relation that does not directly match question keywords. When ODYSSEY prunes up to 53% of graph elements, it severs these factual links. Under our strict context-grounding protocol, models cannot fall back to memorized parametric knowledge; this evidence loss forces the model to emit empty responses (7.20% on 20B, 5.60% on Qwen 30B, and reaching up to 39.0% on constrained slices) and lowers EM to 52.40%–56.80%. Path-Text Top-N hard filtering suffers an even steeper drop (EM of 21.20%–29.53%). In contrast, disabling prompt compression causes EM to collapse by -27.60% (down to 31.60%) while empty answers surge to 52.40% due to severe token blowout. OUEG-RAG avoids this by retaining 100% of the retrieved evidence via compact lossless compression. (b) *Bounded two-tier efficiency*: Rather than relying on unbounded multi-turn agent loops (e.g., HydraRAG), OUEG-RAG concludes unambiguous queries in a single primary call (Fast Exit), reserving a targeted second call only when diagnostic feedback identifies structural collisions or missing passages, where the minimal state capsule provides complete factual sufficiency (Proposition 2). Consequently, OUEG-RAG averages only 1.20–1.22 LLM calls per query, achieving higher accuracy while bounding inference latency.

Systematic Component Ablation Analysis. Table 2 systematically isolates each architectural pillar across 500 pooled evaluation questions on GPT-OSS 120B: (1) *Prompt Compression is Critical*: Disabling lossless topological compression collapses EM by -27.60% (from 59.20% to 31.60%) and F1 by -30.70%, causing over half (52.40%) of all queries to fail with empty answers due to context saturation; (2) *Record Centroids Prevent Row Conflation*: Eliminating row-binding record nodes ρ_i drops EM by -3.00% and F1 by -5.32%, nearly doubling empty abstentions ($11.40\% \rightarrow 19.60\%$) as cell-to-row integrity is lost; (3) *Targeted L2 Evidence Augmentation*: Forbidding new text retrieval during L2 verification drops EM by -3.20% and F1 by -4.46%, with empty answers rising to 18.40%, confirming that targeted passage expansion resolves evidence gaps that re-reading cannot fix; (4) *Diagnostic Focus Resolution*: Omitting fine-grained grounding diagnostics degrades F1 by -2.32% through increased entity normalization er-

Table 2. Systematic knockout ablation study on pooled development partitions ($N = 500$ across Shards 01, 02, and 03 under GPT-OSS 120B). All variants isolate individual architectural components against full OUEG-RAG.

Ablation Configuration	EM (%)	F1 (%)	ΔEM	Empty	Architectural Implication
Full OUEG-RAG (Two-Tier)	**59.20**	**68.06**	—	**11.40%**	Full controlled evidence scaffolding
(1) w/o Prompt Compression	31.60	37.36	−27.60	52.40%	Token blowout; > 52% empty abstentions
(2) w/o Record Nodes ρ_i	56.20	62.74	−3.00	19.60%	Cross-row conflation; doubles empty rate
(3) w/o L2 Targeted Evidence	56.00	63.60	−3.20	18.40%	Re-reading misses external bridge facts
(4) w/o Diagnostic Focus	57.80	65.74	−1.40	15.60%	Lack of cell-level grounding harms F1
(5) w/o Adaptive L2 (Single L1)	58.60	66.78	−0.60	5.40%	L1 fails on collisions (−7.68% F1 on S2)
(6) w/o Path Guidance (Graph)	61.60	70.14	+2.40	11.80%	Flexible reach via multigraph topology

rors; and (5) *Adaptive L2 Verification*: Omitting L2 verification reduces overall F1 (from 68.06% to 66.78%) and severely harms performance on challenging multi-hop splits (dropping F1 from 69.53% to 61.85%, a −7.68% drop on Shard 02), validating L2's role as an essential safeguard for complex collision queries within a bounded 1.22 average call budget.

Human Audit of Provenance Traces ($N = 100$). A manual audit on 100 randomly sampled evaluation questions confirms: (1) *Trace Faithfulness*: In 92.0% of cases, OUEG-RAG produces faithful reasoning chains mapping directly to verified cell coordinates and source passages; (2) *Bridge Alignment Precision*: In 89.0% of cases, the multigraph identifies the ground-truth bridge entity linking the candidate row to external text; and (3) *Empty Failure Reduction*: On 120B, OUEG-RAG reduces empty failures to 0.53% (vs. 3.40% for ODYSSEY and 2.80% for Path-Text).

6 Limitations

First, query-time graph construction and structured prompts incur modest latency over direct prompting ($\approx$ 23–25 s vs. $\approx$ 6–10 s), though significantly lower than multi-turn agent loops. Second, path scoring weights were set via domain heuristics; learned scoring calibration represents a promising future direction. Third, our evaluation focuses on the controlled 1,500-example benchmark for open-weights reproducibility; submitting to blinded leaderboards remains an ongoing milestone.

7 Conclusion

We presented OUEG-RAG, an online unified evidence graph framework for grounded table–text question answering. By uniting schema-preserving multigraphs with compact prompt compression, deterministic path scaffolding, and an adaptive two-tier diagnostic-driven execution protocol, OUEG-RAG resolves the structural trade-off between destructive graph pruning and unbounded agentic loops. Under a strict context-grounded protocol that penalizes unfaithful parametric guessing, empirical evaluations across open-weights models demonstrate that OUEG-RAG consistently achieves higher grounding accuracy, reduces ungrounded empty responses, and produces auditable provenance traces within a bounded 1-to-2 call budget. These findings establish that preserving complete

evidence while providing explicit path guidance offers an effective, training-free paradigm for faithful table–text reasoning.

References

1. Agarwal, A., S, G., Devaguptapu, C.: Hybrid graphs for table-and-text based question answering using LLMs. In: Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2025) (2025)
2. Chen, K., Shen, Y., Zhao, Y., Zhang, W., Zhang, Q.: MuGER2: Multi-granularity evidence retrieval and reasoning for hybrid question answering. In: Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing. pp. 7095–7107 (2022)
3. Chen, S.A., Miculicich, L., Eisenschlos, J.M., Wang, Z., Wang, Z., Chen, Y., Fujii, Y., Lin, H.T., Lee, C.Y., Pfister, T.: TableRAG: Million-token table understanding with language models. In: Advances in Neural Information Processing Systems 37. pp. 21001–21028 (2024)
4. Chen, W., Zha, H., Chen, Z., Xiong, W., Wang, H., Wang, W.Y.: HybridQA: A dataset of multi-hop question answering over tabular and textual data. In: Findings of the Association for Computational Linguistics: EMNLP 2020. pp. 1026–1036. Association for Computational Linguistics (2020)
5. Cheng, Z., Gao, T., Yan, Z., Wang, S., Ye, Z., Guan, Z., Liu, P., Yin, P.: Binding language models in symbolic languages. In: International Conference on Learning Representations (2023)
6. Edge, D., Trinh, H., Cen, N., Cheng, Y., Bradley, R., Chao, A., Mody, A., Truitt, S., Larson, J.: From local to global: A graph RAG approach to query-focused summarization. arXiv preprint arXiv:2404.16130 (2024)
7. Eisenschlos, J.M., Gor, M., Müller, T., Andor, D.: MATE: Multi-view attention for table-text encoding. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics. pp. 7405–7419 (2021)
8. He, X., Tian, Y., Sun, N., Chawla, N.V., Laurent, T., Bresson, X.: G-Retriever: Retrieval-augmented generation for textual graph understanding and question answering. In: Advances in Neural Information Processing Systems 37. pp. 12345–12368 (2024)
9. Lei, F., He, Q., Yang, Y., Hu, B., Zhang, M.: S3HQA: A structure-aware sequential sampling approach for HybridQA. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics. pp. 14021–14035 (2023)
10. Peng, C., Shen, Y., Zhang, M., Chen, J., Han, X., Wang, H.: Graph retrieval-augmented generation: A survey. arXiv preprint arXiv:2408.08921 (2024)
11. Sun, H., Dhingra, B., Srivastava, G., Sun, Y., Cohen, W.W.: DocHopper: A multi-hop question answering system over document graphs. In: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing. pp. 10543–10557 (2021)
12. Sun, Y., Cheng, Q., Zhang, W., Wang, Z.: JointLK: Joint reasoning with language models and knowledge graphs for commonsense question answering. In: Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing. pp. 4402–4415 (2022)
13. Tan, X., Wang, X., Liu, Q., Xu, X., Yuan, X., Zhu, L., Zhang, W.: HydraRAG: Structured cross-source enhanced large language model reasoning. In: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing. pp. 14431–14459 (2025)
14. Wang, Z., Zhang, H., Li, C.L., Eisenschlos, J.M., Perot, V., Lu, Z., Chang, M.W., Lee, C.Y., Pfister, T.: Chain-of-table: Evolving tables in the reasoning chain for table understanding. In: International Conference on Learning Representations (2024)
15. Yasunaga, M., Ren, H., Bosselut, A., Liang, P., Leskovec, J.: QA-GNN: Reasoning with language models and knowledge graphs for question answering. In: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. pp. 535–546 (2021)
16. Ye, Y., Hui, B., Yang, M., Li, B., Huang, F., Li, Y.: Large language models are versatile decomposers: Decompose evidence and questions for table-based reasoning. In: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 174–184 (2023)
17. Yu, X., Jian, P., Chen, C.: TableRAG: A retrieval augmented generation framework for heterogeneous document reasoning. In: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing. pp. 7100–7115 (2025)
18. Zhou, W., Ma, B., Friedrich, A., Mesgar, M.: Table question answering in the era of large language models: A comprehensive survey of tasks, methods, and evaluation. In: Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026) (2026)

Appendix: Proofs of Theoretical Propositions

A. Proof of Proposition 1 (Row Attribute Non-Conflation)

Proof. Let $T = (H, R)$ be a tabular relation with columns $H = (H_1, \ldots, H_m)$ and rows $R = \{r_1, \ldots, r_n\}$. By the graph formulation in Section 3 and Equation (2), each row $r_i \in R$ is assigned a unique centroid vertex $\rho_i \in V_{\text{rec}}$, where $i \in \{1, \ldots, n\}$. Thus, $\rho_i \neq \rho_{i'}$ for any pair of distinct rows $r_i, r_{i'} \in R$ $(i \neq i')$.

The star-topology subgraph $\mathcal{G}^*(\rho_i) = (V_i, E_i)$ has vertex set:

$$V(\mathcal{G}^*(\rho_i)) = \{\rho_i, u_i\} \cup \{v_{i,j}\}_{j=1}^{m}. \tag{14}$$

By construction, the intersection of $V(\mathcal{G}^*(\rho_i))$ with the record node set V_{rec} contains strictly the single centroid node ρ_i:

$$V(\mathcal{G}^*(\rho_i)) \cap V_{\text{rec}} = \{\rho_i\}. \tag{15}$$

Consequently, for any two distinct rows $r_i, r_{i'} \in R$ $(i \neq i')$, their respective star centroids are disjoint:

$$V(\mathcal{G}^*(\rho_i)) \cap V(\mathcal{G}^*(\rho_{i'})) \cap V_{\text{rec}} = \{\rho_i\} \cap \{\rho_{i'}\} = \emptyset. \tag{16}$$

Now consider an arbitrary cell value $v_{i,j}$ originating from row r_i. In $\mathcal{G}^*(\rho_i)$, the directed incident edge from V_{rec} to $v_{i,j}$ is uniquely given by $(\rho_i, \text{norm}(H_j), v_{i,j})$. Even in the event that row $r_{i'}$ contains an identical literal cell value (i.e., $v_{i,j} = v_{i',j'} = v$), the schema-preserving multigraph assigns separate directed edges from their distinct row centroids:

$$(\rho_i, \text{norm}(H_j), v) \in E(\mathcal{G}^*(\rho_i)), \quad (\rho_{i'}, \text{norm}(H_{j'}), v) \in E(\mathcal{G}^*(\rho_{i'})). \tag{17}$$

Because no direct edge connects ρ_i and $\rho_{i'}$, the shortest path between $v_{i,j}$ and any attribute value $v_{i',k}$ belonging to a different row $r_{i'}$ is at least 2 hops (traversing through a shared literal node or external passage entity). Within any 1-hop star traversal from ρ_i, every attribute value $v_{i,j}$ is uniquely and unambiguously bound to its parent row r_i under predicate $\text{norm}(H_j)$, mathematically precluding cross-row attribute conflation. □

B. Proof of Proposition 2 (Informational Sufficiency of $\mathcal{K}_{\text{L2}}$)

Proof. Let $\mathcal{X}_{\text{L1}}$ denote the complete input evidence available at Tier 1 (L1), consisting of the query q, candidate row subgraphs $\{\mathcal{G}^*(\rho_i)\}_{i=1}^{k}$, and retrieved passage texts $\mathcal{D}_{\text{L1}}$. Let a^* denote the target answer to query q.

Suppose Tier 1 evaluation produces an unresolved diagnostic check ψ_{check} and a focus candidate set $\mathcal{C}_{\text{focus}} \subset R$. By the definition of the targeted diagnostic verification in Equation (10), any discarded row $r \in R \setminus \mathcal{C}_{\text{focus}}$ is deterministically refuted by L1 (i.e., satisfies $\neg\psi_{\text{cand}}(r)$ under verified cell constraints), which implies:

$$P(a^* \in \text{Attr}(r) \mid q, \psi_{\text{check}}) = 0 \quad \forall r \notin \mathcal{C}_{\text{focus}}. \tag{18}$$

Hence, the support of the conditional distribution of a^* is strictly restricted to the attributes of the focus candidate set $\mathcal{C}_{\text{focus}}$ and their linked external passages:

$$a^* \in \bigcup_{c \in \mathcal{C}_{\text{focus}}} \left(\text{Attr}(c) \cup \text{Ent}(\mathcal{D}_c)\right). \tag{19}$$

Let $\mathcal{X}_{\text{discard}} = \mathcal{X}_{\text{L1}} \setminus (\{q, \psi_{\text{check}}\} \cup \{\mathcal{G}^*(\rho_c)\}_{c \in \mathcal{C}_{\text{focus}}})$ denote the discarded context comprising non-candidate rows and unlinked passages. Because a^* is conditionally independent of $\mathcal{X}_{\text{discard}}$ given the true candidate partition $\mathcal{C}_{\text{focus}}$ and the resolving supplemental text $\Delta\mathcal{D}$ that grounds ψ_{check}, we have:

$$P(a^* \mid \mathcal{X}_{\text{L1}} \cup \Delta\mathcal{D}, \mathcal{K}_{\text{L2}}) = P(a^* \mid \mathcal{K}_{\text{L2}}), \tag{20}$$

where $\mathcal{K}_{\text{L2}} = (q,\, \psi_{\text{check}},\, \{\mathcal{G}^*(\rho_c)\}_{c \in \mathcal{C}_{\text{focus}}},\, \Delta\mathcal{D})$ as defined in Equation (11).

By the definition of conditional mutual information:

$$\begin{aligned} I(a^*; \mathcal{X}_{\text{L1}} \cup \Delta\mathcal{D} \mid \mathcal{K}_{\text{L2}}) &= H(a^* \mid \mathcal{K}_{\text{L2}}) - H(a^* \mid \mathcal{K}_{\text{L2}}, \mathcal{X}_{\text{L1}} \cup \Delta\mathcal{D}) \\ &= H(a^* \mid \mathcal{K}_{\text{L2}}) - H(a^* \mid \mathcal{K}_{\text{L2}}) = 0. \end{aligned} \tag{21}$$

Therefore, the state capsule $\mathcal{K}_{\text{L2}}$ preserves all answer-bearing mutual information, forming a sufficient statistic for a^* while pruning irrelevant non-candidate evidence. □